Sangsidhya Kar

Kolkata, West Bengal | sangsidhyakar@gmail.com | sangsidhya.github.io

Education

Presidency University — Kolkata, WB
B.Sc (Honours with Research) in Statistics — *Expected 2029*

St. Mary's Orphanage and Day School — Kolkata, WB
Indian School Certificate (Class 12) – 94.25% — *Graduated 2025*

Research Experience

Global Games with Heavy Tails (*ssrn: 6296258*)
S. Kar

A Dynamic Central Limit Theorem for Mean-Field Quantum Filtering (*arXiv: 2609.13346*)
***S. Kar**, V. N. Kolokoltsov*

When Known Objects Hide Unknowns: Sparse-Contamination Theory for CLIP-Based OOD Detection (Under review)
D. Liu, ***S. Kar**, R. Piao, M. Burenko, W. Chang*

Kernel Scaling in Feature Learning Representations (Under review)
***S. Kar**, M. Sangal, W. Chang, D. Kim, B. Li*

Decentralized Learning of Stackelberg Equilibrium in General-Sum Leader-Controller Markov Games (Under review)
***S. Kar**, B. Chang, J. Loi, W. Chang, Á. Parafita*

Why Adaptive Optimizers Underestimate Rare Tokens (*arXiv: 2609.37535*)
S. Kar

Honest Finite-sample Inference for Wasserstein Distances in General Dimension *(ssrn: 7620318)*
S. Kar

Proof Fragility Prediction: Step-Level Failure Localization for Efficient Neural Theorem Proving (Under review)
*Z. Li, R. Sankhyan, **S. Kar**, W. Chang*

Projects & Miscellaneous

Developed multiple productivity applications, including a Pomodoro timer, a thermodynamic system modeling app, a task management system, a PDF reader with advanced features, and an academic grade tracker.

Most apps can be found at https://github.com/Sangsidhya/Sangsidhya.github.io.

Operator Algebras and Quantum Theory (Notes)

My article *The Topography of Silent Thoughts* published in Elements.

Certifications & Achievements

Active member of local swimming club (2013-2023); won multiple accolades in competitive swimming

Was selected for ISI LIMIT 2025 and MTRP 2024 camps; secured 3rd rank in both IISc Enumeration 2025 and the C. R. Rao Statistics Olympiad. Also Selected in Madhava camp held through 22nd to 26th June, 2026.

Interests

Piano, Chess, Coding, Sports

Last updated: October 17, 2026